Continual Learning for Adaptive Health and Reuse Assessment of Heterogeneous Second-Life EV Batteries

Tarun Sitaraman
Department of Networking and Communications
SRM Institute of Science and Technology
Chennai, India
RA2411028010038

Sohan Sandeep Krishnan
Department of Networking and Communications
SRM Institute of Science and Technology
Chennai, India
RA2411028010014

Darshan D
Department of Networking and Communications
SRM Institute of Science and Technology
Chennai, India
RA2411028010056

***Abstract*—Retired electric-vehicle batteries can serve in second-life applications, but a health model trained on one battery population may fail on the next, and continual learning promises to adapt without forgetting. We evaluate this promise for forecasting the state of health (SOH) of second-life lithium-ion cells one to three capacity checks ahead. From five public datasets (395 cells after quality rules) we build a leakage-controlled benchmark: every file is verified against its repository record, labels are reproduced from raw data, and two label artefacts are removed, test currents set from the previous capacity and pulse charge counted in the capacity. A Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) is compared with fine-tuning, replay, joint training and simple forecasters under three protocols: a continual stream of cells or usage conditions, unseen groups, and pretraining on 369 first-life cells of other chemistries. Learned forecasters beat persistence, and forgetting appears whenever usage conditions differ (0.18–0.46 percentage points). Trained directly, TFT + EWC is never the most accurate method, and first-life pretraining does not help. Trained instead on the correction to linear extrapolation of the recent fade rate, it becomes the most accurate learned forecaster on the S6 stream, 0.17–0.30 percentage points better than its direct form, level with extrapolation for horizons of one to three checks, and EWC then reduces forgetting over six horizons; none of the models beats extrapolation. Inputs from the 1.4 billion ageing measurements between checks, processed with Apache Spark, make the forecasts worse. A re-run shows that several significance results depend on the software environment, so the main results are required to replicate on a second seed set. Forecast SOH maps to reuse classes that agree with the measured class for 95–97% of samples on unseen usage conditions.**

***Index Terms*—Continual learning, second-life batteries, electric vehicles, state of health, battery degradation, heterogeneous batteries, catastrophic forgetting, machine learning**

I. Introduction

Lithium-ion batteries lose capacity and power with cycling, temperature and time. When an electric-vehicle (EV) battery no longer meets automotive requirements, it may still serve less demanding applications such as stationary storage, renewable-energy buffering or backup power. Reuse extends the battery's life and delays recycling, but only if its health is assessed reliably: a battery that suits one application may not suit another.

The state of health (SOH) describes the present capability of a battery relative to a reference condition, and machine-learning models have estimated it from voltage, current, temperature, partial charging and impedance measurements [1]–[4]. Retired batteries are, however, heterogeneous: chemistry, manufacturer, capacity, first-life history, temperature and charging protocol differ, and complete first-life records are often missing. A model trained once on a fixed population may therefore fail on the next population. Continual learning addresses this by updating a model on new data while preserving what it learned earlier; it must therefore be evaluated both on the newly introduced battery group and on the groups learned before, to measure catastrophic forgetting.

This work combines continual learning, health assessment and second-life reuse assessment. Health is assessed by *forecasting* the SOH one to three capacity checks ahead from everything measured up to the latest check: on the first dataset studied, estimating the current SOH from cycling features was beaten by carrying the last measured SOH forward (§III), whereas a reuse decision needs the future SOH. The forecast is mapped to reuse-suitability classes.

The contributions are:

- A leakage-controlled benchmark for forecasting the SOH of second-life cells, built from five public datasets (395 cells after quality rules) verified against their repository records, with automated data checks and protocols fixed before model training.
- Two label artefacts identified and removed: test currents set from the previous capacity, which encode the label, and pulse charge counted in the capacity, which shifts labels by up to 0.09 percentage points.
- An evaluation of a Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) against fine-tuning, replay, joint training and simple forecasters under three protocols (a continual stream, unseen groups and

first-life pretraining), reporting adaptation, forgetting and retention.

- A continual TFT + EWC that learns the correction to linear extrapolation: it is the most accurate learned forecaster on the S6 stream, level with extrapolation for horizons 1–3, and its EWC reduces forgetting over longer horizons (replicated).
- A reproducibility analysis showing that several significance results on small battery datasets depend on the numerical environment, and a replication rule that the main results satisfy.

II. Related Work

The review covers 2022–2026 studies on second-life lithium-ion batteries, SOH estimation, continual learning and reuse assessment (Table I).

SOH estimation for second life. Braco *et al.* estimated the SOH of second-life cells from partial charging under residential and fast-charging-station profiles, with offline RMSE of about 0.6 and 0.8% and 85% less testing time than a full capacity test [1]. In a companion study on six reused Nissan Leaf modules they compared 58 health indicators from incremental-capacity analysis, partial charging, constant-current/constant-voltage stages and internal resistance; the best indicator at repurposing reached about 0.2% RMSE, but the most useful indicators changed as ageing progressed [2]. Faraji-Niri *et al.* accelerated second-life SOH estimation with electrochemical impedance spectroscopy (EIS) and machine learning [3]; EIS is fast and non-destructive but sensitive to temperature, state of charge and equipment. Takahashi *et al.* screened batteries for second life with statistical current and voltage features and bagged Gaussian process regression, with root-mean-squared percent error below 1.48% in the worst cases [4]. Cui *et al.* tested retired Nissan Leaf cells for more than 15 months under grid-storage-like operation and estimated health online with clustering, with mean absolute percentage error below 2.3% [5]. Vignesh *et al.* reviewed SOH estimation methods for second-life batteries and their challenges [6]. These studies estimate the *current* SOH of a fixed population; none adapts a model as new, different populations arrive.

Continual learning and generalisation. Lee *et al.* applied bilevel-optimised continual learning to predict the capacity degradation of lithium-ion batteries [7]. Yao *et al.* used progressive neural networks with temporal convolutional networks for online state-of-charge estimation across three public drive-cycle datasets, explicitly addressing catastrophic forgetting and task order [8]; the target, however, is state of charge, not SOH or reuse. Li *et al.* proposed domain-generalised SOH estimation for second-life EV batteries [9], and Meskine *et al.* compared BiLSTM, TCN and probabilistic TCN-GPR models on sparse, noisy data, where isotonic recalibration substantially reduced the calibration error [10]. Transfer and domain generalisation do not by themselves address a long sequence of tasks with explicit retention.

Research gap. Second-life SOH estimation, screening, transfer learning and degradation prediction have progressed largely separately; continual learning is rarely combined with heterogeneous second-life SOH forecasting, an explicit reuse decision and an evaluation of forgetting. This work asks whether a continual learner can learn heterogeneous second-life populations in sequence while retaining earlier ones, and whether its forecasts support reuse decisions.

TABLE I
Summary of Reviewed Literature

Study	Methodology	Main Challenge
Braco 2022 [1]	Partial-charge / real-profile SOH	Depends on charging profile
Braco 2022 [2]	Health-indicator selection	Indicators change with ageing
Faraji-Niri 2023 [3]	EIS + machine learning	Measurement conditions
Takahashi 2023 [4]	Features + bagged GPR	Feature/data quality
Cui 2024 [5]	Online ML + clustering	Limited experimental population
Vignesh 2024 [6]	SOH methods review	No single adaptive model
Lee 2024 [7]	Bilevel-optimised CL	Adaptation versus retention
Yao 2025 [8]	PNN + TCN continual learning	Focused on SOC
Li 2025 [9]	Domain generalisation	Not continual learning
Meskine 2026 [10]	BiLSTM/TCN + probabilistic model	Sparse/noisy data

III. Methodology

Given everything measured up to a reference performance test (RPT, or capacity check), we forecast the SOH one to three checks ahead. Cells or usage conditions arrive as a stream of tasks, and a model must adapt to each new task without forgetting the earlier ones. Fig. 1 summarises the framework.

Change of task after inspecting the data. The study was first designed to estimate the *current* SOH from cycle-level features. On the first dataset these features carried almost no health information (median within-cell Spearman correlation with SOH 0.09, against -0.99 for elapsed time), and learned estimators were worse than persistence (1.78 and 2.37 pp for gradient boosting and ridge regression, against 0.99 pp). We therefore reframed the task as forecasting, after seeing the data; the forecasting protocol was then committed before any forecasting model was fitted, and deviations are reported.

A. Datasets

Table II lists the five public datasets (all CC BY 4.0). Every file is verified against its repository record or checksums, and our parsing of the raw files reproduces exactly the RPT capacities of the processed S1 tables and the authors' capacities of all 843 S6 checks.

S1 (Stanford) contains six LG Chem INR21700-M50T cells that spent 23 months under an EV driving profile with different charge rates [21] and then about 27 months on synthetic peak-shaving duty cycles [22]; each RPT is a C/20 discharge interrupted by C/2 pulses at three voltages, with impedance spectroscopy (EIS) from the fifth RPT [11]. Its heterogeneity lies in first-life ageing and duty cycle among cells of one type. Data-quality findings are handled by fixed rules: RPT 2 (run at

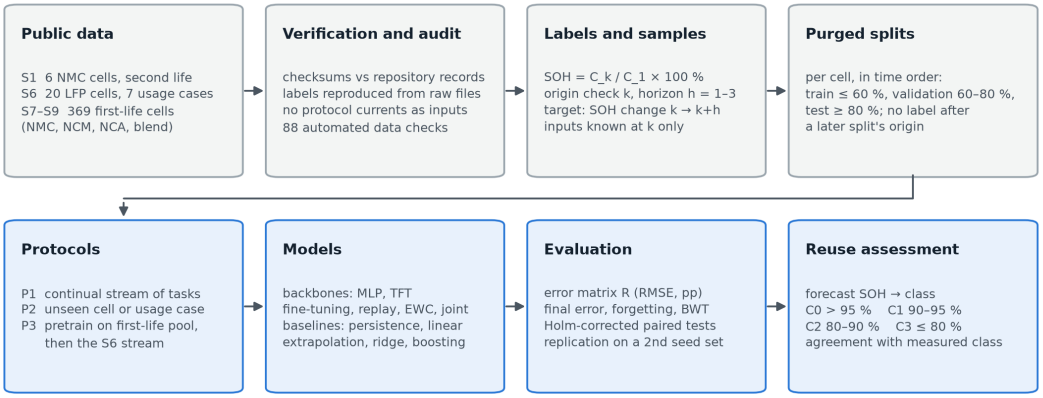


Fig. 1. Framework of the study. Public data are verified and audited, converted into forecasting samples with purged chronological splits, and evaluated under three protocols with continual-learning metrics, a replication rule and a reuse-class mapping.

TABLE II
DATASETS. S1 AND S6 FORM THE CONTINUAL-LEARNING STREAMS; S7–S9 FORM THE FIRST-LIFE PRETRAINING POOL.

Set	Cells and conditions	Label	Samples
S1 [11], [12]	6 NMC 21700 cells, second life on residential or commercial peak shaving, 20/35 °C blocks	C/20 RPT capacity, pulses excluded	279
S6 [13], [14]	20 LFP 18650 cells, 7 conditions (charge mode, voltage and rate, discharge, depth), 50 °C	C/25 check capacity	2,409
S7 [15], [16]	195 of 228 NMC/C+SiO cells, calendar, cyclic and WLTP ageing, 0–40 °C	check-up capacity	10,251
S8 [17], [18]	55 NCM cells, six charge/discharge protocols	capacity cycles	9,699
S9 [19], [20]	119 of 130 NCA, NCM and blended cells, 25–45 °C	capacity cycles	16,761

about 20 °C) is excluded, off-protocol pulses and galvanostatic spectra are set to missing, and RPT 7's aggregate EIS files are decoded and checked against the authors' values.

S6 (Wheeler *et al.*) was selected by criteria fixed before screening: a reuse licence, a series of capacity checks, ageing past first-life end of life, heterogeneity that S1 lacks and verifiable metadata. Its cells were characterised every 100 cycles and end at 32–64% SOH; "second life" here means continued ageing below 80% SOH under the same protocol. A check is used only if its C/25 discharge starts from full charge and reaches the cut-off (843 of 847 checks).

S7–S9 are first-life datasets used for pretraining. Cells with at least five valid checks are kept: 369 cells and 36,711 samples.

B. Big-data processing of the ageing cycles

Between checks, every S6 cell logged each ageing cycle at about one sample per second: 827 files (one per cell and 100-cycle block), 132 GB raw and about 1.4×10^9 rows. They are reduced to block-level inputs with Apache Spark 4.2 (PySpark) in local mode (Fig. 2). In the map stage, one Spark task per block parses the raw rows into steps and derives, for every cycle, a validity flag and seven measured responses: end-of-charge and end-of-discharge voltage, constant-voltage hold time and final current, relaxation over the rest after charge and after discharge, and internal resistance where recorded. Charge, energy and current are excluded because the protocol scales them with the previous capacity. The per-cycle table (about 83,000 cycles) is cached, and Spark SQL then groups the valid cycles by cell and block into the mean, last value and least-squares slope of each response (21 inputs), requiring at least 50 valid cycles. The block ending at a check joins the origin at that check; every block is verified to end before it. On six worker cores, the full run (827 map tasks, 82,529 cycles, 97% valid) takes 20 min. Its output matches an independent pandas implementation on all 827 blocks within 10^{-9} relative plus 10^{-12} absolute (largest difference 2×10^{-13}); a purely relative criterion, stated first, failed only on a slope that is zero up to rounding. Discharged charge, which is excluded, correlates at $r = 0.97$–0.999 within cell with the previous check's capacity, confirming the protocol scaling.

C. State of health and forecasting target

For cell b with check capacities $C_{b,1}, C_{b,2}, \ldots$ in time order,

$$\mathrm{SOH}_{b,k} = 100\, C_{b,k}/C_{b,1}\ \%, \tag{1}$$

i.e. relative to the first valid measurement, not to a nominal rating; for S1 this is already an aged state. In S1 the capacity is the charge of the C/20 steps only [11, Eq. (1)]. The processed tables we started from also counted the pulse charge, which shifts some labels by up to 0.09 pp, comparable to one RPT's fade.

A sample is a triple (cell b, origin k, horizon $h \in \{1, 2, 3\}$), about one to three months in S1 and 100–300 cycles in S6. The target is

$$\Delta_{b,k,h} = \mathrm{SOH}_{b,k+h} - \mathrm{SOH}_{b,k}, \tag{2}$$

so an error in Δ equals the error in the future SOH; errors are in SOH percentage points (pp). Over a set $\mathcal{D}$ of N samples,

$$\mathrm{RMSE}(\mathcal{D}) = \sqrt{\tfrac{1}{N} \textstyle\sum_{\mathcal{D}} \left(\hat{\Delta}_{b,k,h} - \Delta_{b,k,h}\right)^2}. \tag{3}$$

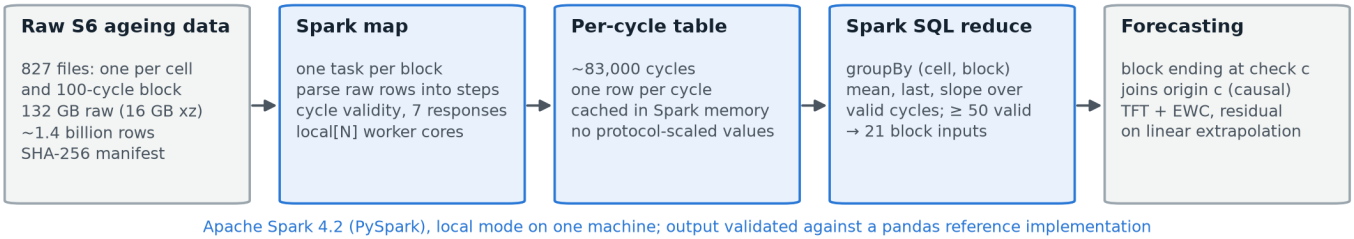


Fig. 2. Big-data pipeline for the S6 ageing cycles: Spark map (one task per 100-cycle block) to a cached per-cycle table, Spark SQL reduction to block inputs, and a causal join to the forecasting origins.

The fade rate $r_{b,k}$ is the least-squares slope of SOH against time t (days or cycles) over the last three valid checks,

$$r_{b,k} = \frac{\sum_{i=k-2}^{k}(t_{b,i} - \bar{t})(\mathrm{SOH}_{b,i} - \overline{\mathrm{SOH}})}{\sum_{i=k-2}^{k}(t_{b,i} - \bar{t})^2}. \tag{4}$$

Inputs use only information available at the origin. **F0** contains the SOH history ($\mathrm{SOH}_{b,k}$, time or cycles since the first check, $r_{b,k}$), static descriptors (S1: duty cycle and first-life history [11]; S6: test conditions) and the *planned* usage until the target (S1: days at each temperature set point and at rest; S6: cycles), so forecasts are conditional on the planned usage. **F1** adds diagnostics measured at the origin: 47 in S1 (pulse resistances, charge between pulse voltages, incremental-capacity peaks of the C/20 discharge and EIS features) and 17 in S6 (plateau charges, incremental-capacity peaks, hysteresis, a resistance proxy and capacity ratios). No test current is an input: in S1 the RPT and monthly discharge currents were set from the previous capacity and therefore encode the label. Missing values are imputed from training data and flagged.

D. Splits and protocols

Purged chronological splits. For each cell, cut A lies at 60% and cut B at 80% of its valid checks. Training samples have their target at or before A, validation samples their origin at or after A and target at or before B, and test samples their origin at or after B; samples straddling a cut are dropped, so no training label lies after a validation origin and no validation label after a test origin.

P1, continual stream. A task is a cell in S1 (six tasks) or a test condition in S6 (seven tasks). At step i of an ordering π, a strategy receives only task π_i's data. Each method runs on 10 orders with 5 seeds, identical across methods.

P2, unseen group. One cell (S1) or condition (S6) is held out; models are trained on the others and tested on all its samples.

P3, first-life pretraining. All datasets are mapped onto shared inputs: SOH history, calendar and cycle time, planned time to the target and operating conditions (temperature, charge and discharge rates, depth of discharge, charge voltage and CV phase, storage SoC, chemistry), with undefined conditions flagged as missing. Models are pretrained on the pool, which also supplies the input scaling and the first EWC anchor; the seven S6 conditions then follow as a stream. We compare training from scratch, pretraining with fine-tuning and pretraining with EWC for both backbones, and record the pool test error before and after the stream.

E. Models

Preprocessing (standardisation, median imputation with missing indicators, target scaling) is fitted on the first task and then frozen. The *MLP* has hidden layers of 128 and 64 ReLU units (dropout 0.1). The proposed backbone adapts the Temporal Fusion Transformer (TFT) [23]: a window of the $W{=}4$ most recent origins of the same cell and horizon (inputs only, never labels) passes through per-variable embeddings, variable selection by gated residual networks, an LSTM encoder [24] and interpretable two-head attention masked to past steps.

Strategies. Fine-tuning trains on the current task only (lower bound); experience replay [25] stores B samples per finished task; joint training fits all tasks at once (upper bound); and Elastic Weight Consolidation (EWC) [26] adds, for each finished task j,

$$\frac{\lambda}{2}\sum_{p} F_{j,p}\left(\theta_p - \theta^*_{j,p}\right)^2, \tag{5}$$

with θ^*_j the parameters after task j and F_j the empirical diagonal Fisher information over up to 500 training samples. *TFT + EWC* is the proposed model. Training uses Adam [27], batches of 64, at most 200 epochs and early stopping on the current task's validation RMSE.

Reference forecasters are persistence ($\hat{\Delta} = 0$), linear extrapolation $\hat{\Delta}_{b,k,h} = r_{b,k}(t_{b,k+h} - t_{b,k})$, ridge regression (on planned usage only, F0 or F1) and histogram gradient boosting on F1.

Learned residual. Because linear extrapolation proved hardest to beat, a follow-up experiment, specified and committed before it was run, trains the continual learners on the correction to it. With $\hat{\Delta}^{\mathrm{lin}}_{b,k,h}$ the extrapolation above (zero when $r_{b,k}$ is missing), the model g_θ learns $\Delta_{b,k,h} - \hat{\Delta}^{\mathrm{lin}}_{b,k,h}$ and the forecast is

$$\hat{\Delta}_{b,k,h} = \hat{\Delta}^{\mathrm{lin}}_{b,k,h} + g_\theta(\mathbf{x}_{b,k,h}). \tag{6}$$

Since $\hat{\Delta}^{\mathrm{lin}}$ is known at the origin, every error is unchanged by this reparameterisation. The residual and the direct (target Δ) forms are compared on the S6 stream with the P3 shared inputs, from scratch, for horizons 1–3 (E1) and 1–6 (E2), each tuned and evaluated as P3.

F. Hyperparameters, metrics and statistics

Hyperparameters are chosen on validation data from tuning orders disjoint from the evaluation orders: the learning rate per backbone (lowest validation plasticity $\frac{1}{T}\sum_j R_{j,j}$ of fine-tuning streams), then replay $B \in \{50, \dots, 400\}$ and EWC $\lambda \in \{10^0, \dots, 10^5\}$; a selection at a grid edge extends the grid. For P3, ten tuning orders are used with the one-standard-error rule

$$g^\star = \min\left\{\, g : \bar{m}_g \le \bar{m}_{\hat{g}} + s_{\hat{g}} \right\}, \quad \hat{g} = \arg\min_g \bar{m}_g, \qquad (7)$$

where $\bar{m}_g$ is the mean validation criterion of grid value g and s_g its standard error.

After each step i, every task's test set is evaluated, giving $R_{i,j} = \mathrm{RMSE}(\mathcal{D}^{\mathrm{te}}_{\pi_j} \mid \theta_i)$. Following [28], [29], we report the final average error $\frac{1}{T}\sum_j R_{T,j}$, backward transfer $\frac{1}{T-1}\sum_{j<T}(R_{T,j} - R_{j,j})$ and forgetting $\frac{1}{T-1}\sum_{j<T}(R_{T,j} - \min_{j\le i<T} R_{i,j})$ (positive values mean forgetting), the skill $1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{persistence}}$, and the agreement of the reuse class of the forecast SOH with the measured class (C0 $> 95\%$, C1 90–95%, C2 80–90%, C3 $\le 80\%$, fixed in advance).

Means carry 95% bootstrap intervals [30]. Methods are compared by two-sided Wilcoxon signed-rank tests [31] on matched runs or held-out groups, with rank-biserial effect sizes [32] and Holm correction [33]. A difference d is *replicated* only if it is significant in the main seed set (0–4) and in a replicate set (5–9), with the same sign and $|d| \ge 0.05$ pp in both; $|d| < 0.05$ pp in both means practically equivalent. The threshold is about half the label shift caused by the pulse-charge definition.

G. Implementation and availability

The pipeline uses Python 3.11, NumPy 2.4, pandas 3.0, scikit-learn 1.9 and PyTorch. P1 and P2 were first run on an Intel Xeon (PyTorch version not recorded) and re-run, like P3 and the ablation, with PyTorch 2.14.1 on an Intel Core 5 210H, with one CPU thread and seeded, deterministic algorithms. Every result file records configuration, seeds, library versions and the Git commit. Unit tests cover input causality, split purging and the exclusion of labels and protocol currents from the inputs. The datasets are at doi:10.17605/OSF.IO/8JNR5 (S1), 10.57745/OLBXKT (S6), 10.35097/1969 (S7), 10.5281/zenodo.10963339 (S8) and 10.5281/zenodo.6405084 (S9); code, decision log and results are at https://github.com/TarunSitaraman/continual-learning-heterogeneous-second-life-ev-batteries.
[TODO: Confirm repository visibility and code licence before submission.]

IV. Results

Data. S1 diagnostics track the current SOH (median within-cell $\rho = 0.97$ for the main incremental-capacity peak) but not the change *to the next RPT* (median $|\rho| \le 0.3$), and every S1 sample stays in class C0. S6 conditions fade very differently, and all cells cross from C0 to C3.

A. Continual stream (P1)

Table III gives the final average error, skill and forgetting.

S1. TFT + EWC reaches 0.462 pp (skill 0.27), level with gradient boosting (0.466 pp), TFT fine-tuning (0.467 pp) and MLP replay (0.475 pp); a ridge regression on the planned usage alone is slightly better (0.450 pp). Forgetting is small for every method (0.016–0.040 pp). Persistence and linear extrapolation degrade quickly with the horizon (0.83 and 0.90 pp at h=3), while the learned models stay between 0.42 and 0.49 pp. Comparisons with TFT + EWC whose significance changed in a re-run differ by only 0.005–0.014 pp, an order of magnitude below the label shift of the pulse-charge definition (§IV-C); we therefore report S1 method differences as practically equivalent.

S6. When tasks differ in usage, forgetting is substantial (0.18–0.46 pp). EWC lowers the TFT's mean error and mean forgetting relative to fine-tuning (0.677 against 0.883 pp), but which of the two differences is significant changed between runs (§IV-C), so we do not claim a significant benefit. TFT + EWC is not the most accurate method: MLP fine-tuning (0.531 pp) and MLP + EWC (0.540 pp) are significantly better in both runs, the joint upper bounds (0.414 and 0.427 pp) put the TFT's continual-learning cost at about 0.25 pp, and linear extrapolation of the recent fade rate (0.343 pp) and gradient boosting (0.372 pp) are the best forecasters. The TFT varies strongly with task order (s.d. 0.26 pp with EWC, 0.54 pp with fine-tuning), and the diagnostic features do not help it (0.621 pp without them).

B. Unseen cell or usage condition (P2)

Gradient boosting is best on both datasets (Table IV). On S1 the MLP, the TFT and the usage-only ridge are close (0.285–0.301 pp); on S6 the 100% depth-of-discharge condition is the hardest for every method (1.8–2.4 pp). The reuse class of the forecast SOH agrees with the measured class for 95–97% of S6 samples with the learned models (macro-F1 0.83–0.88), against 85% (0.64) for persistence. With six or seven held-out groups, no difference with the TFT survives Holm correction.

C. Reproducibility of the comparisons

P1 and P2 were re-run in a pinned environment on data rebuilt from the raw archives and verified against recorded checksums. Every baseline, both joint upper bounds, every P2 result and the ranking of methods reproduced; the neural models moved by at most 0.059 pp (TFT + EWC on S6: 0.677 to 0.704 pp). On S1, five Holm-corrected comparisons with TFT + EWC crossed $p = 0.05$. On S6, validation selected EWC $\lambda = 1000$ instead of 100, a 0.008 pp tie over three tuning runs; the error advantage of TFT + EWC over TFT fine-tuning then moved from $p_{\mathrm{Holm}} = 0.005$ to 0.060, while its reduction of forgetting moved from 0.050 to 0.008. These findings motivated the ten tuning orders, the one-standard-error rule and the replication rule used for P3.

TABLE III

PROTOCOL P1 (CONTINUAL STREAM). FINAL AVERAGE TEST RMSE OF SOH($k+h$) IN PP (MEAN ± S.D. OVER 10 ORDERS × 5 SEEDS FOR CONTINUAL METHODS, OVER 5 SEEDS FOR JOINT TRAINING), SKILL RELATIVE TO PERSISTENCE ($1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{persistence}}$) AND FORGETTING (PP). †UPPER BOUND, TRAINED ON ALL TASKS AT ONCE.

Method	S1: six NMC second-life cells RMSE	Skill	Forgetting	S6: 20 LFP cells, seven usage conditions RMSE	Skill	Forgetting
Persistence	0.632	0.00	–	1.644	0.00	–
Linear extrapolation	0.726	-0.15	–	0.343	0.79	–
Ridge, usage only	0.450	0.29	–	1.274	0.22	–
Ridge, F0	0.678	-0.07	–	0.801	0.51	–
Ridge, F1	0.491	0.22	–	0.605	0.63	–
Gradient boosting, F1	0.466	0.26	–	0.372	0.77	–
MLP fine-tuning	0.492 ± 0.029	0.22	0.030	0.531 ± 0.151	0.68	0.207
MLP replay	0.475 ± 0.021	0.25	0.025	0.597 ± 0.178	0.64	0.212
MLP + EWC	0.497 ± 0.033	0.21	0.040	0.540 ± 0.176	0.67	0.212
MLP joint†	0.500 ± 0.016	0.21	–	0.414 ± 0.037	0.75	–
TFT fine-tuning	0.467 ± 0.014	0.26	0.026	0.883 ± 0.538	0.46	0.461
TFT + EWC	0.462 ± 0.019	0.27	0.016	0.677 ± 0.261	0.59	0.274
TFT + EWC, F0	0.472 ± 0.020	0.25	0.019	0.621 ± 0.257	0.62	0.176
TFT joint†	0.455 ± 0.017	0.28	–	0.427 ± 0.049	0.74	–

TABLE IV

PROTOCOL P2 (UNSEEN BATTERY IN S1, UNSEEN USAGE CONDITION IN S6): MEAN TEST RMSE (PP) OVER HELD-OUT GROUPS, RANGE OVER GROUPS, AND FOR S6 REUSE-CLASS ACCURACY / MACRO-F1 OF THE FORECAST SOH.

Data	Method	RMSE	Range	Class acc./F1
S1	Persistence	0.545	0.436–0.688	–
S1	Linear extrapolation	0.511	0.284–0.725	–
S1	Ridge, usage only	0.301	0.172–0.409	–
S1	Ridge, F0	0.418	0.158–0.779	–
S1	Ridge, F1	0.320	0.161–0.477	–
S1	Gradient boosting, F1	0.219	0.091–0.299	–
S1	MLP	0.285	0.149–0.389	–
S1	TFT	0.292	0.199–0.377	–
S6	Persistence	2.817	2.025–4.327	0.85 / 0.64
S6	Linear extrapolation	1.372	0.856–1.669	0.94 / 0.75
S6	Ridge, usage only	1.177	0.870–2.288	0.95 / 0.83
S6	Ridge, F0	0.859	0.598–1.836	0.96 / 0.86
S6	Ridge, F1	1.137	0.588–2.188	0.96 / 0.85
S6	Gradient boosting, F1	0.710	0.395–1.839	0.97 / 0.88
S6	MLP	0.821	0.476–2.225	0.96 / 0.87
S6	TFT	0.868	0.551–2.390	0.96 / 0.88

D. First-life pretraining (P3)

The pool provides 36,711 samples, against 2,409 in S6. Validation selected EWC $\lambda = 1000$ for the MLP but only 0.01 for the pretrained TFT. Under the replication rule (Fig. 3):

- EWC and fine-tuning are practically equivalent for the pretrained TFT in final error, forgetting and backward transfer (differences of 0.015–0.017 pp in both seed sets);
- pretrained TFT + EWC beats pretrained MLP fine-tuning (by 0.334 and 0.185 pp), also in forgetting and backward transfer;
- both backbones are better from scratch on average (TFT 0.401/0.415 pp against 0.509/0.591 pp pretrained with EWC), significant for the TFT only in the replicate set;
- linear extrapolation (0.343 pp) remains the most accurate method.

The S6 stream does not degrade the pool test error

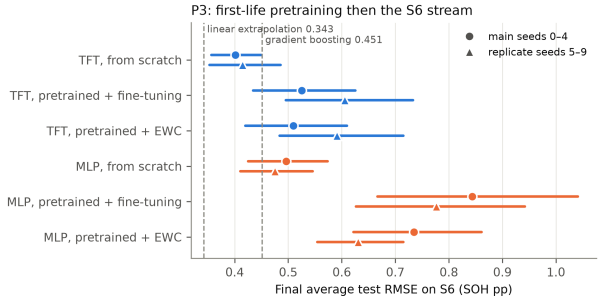


Fig. 3. P3: final average test RMSE on S6, mean and 95% bootstrap CI over 50 runs per seed set. Dashed: linear extrapolation and gradient boosting.

(TFT + EWC: 3.280 to 3.148 pp), but on that split the pretrained models are less accurate than linear extrapolation (3.001 pp): with the shared inputs, pretraining does not learn a first-life model worth transferring. Pretraining takes 22 s (MLP) and 251 s (TFT) on one CPU thread; EWC adds about 50% to the TFT's stream time without improving accuracy.

Why the TFT improved. A post hoc ablation, repeated on the replicate seeds (Fig. 4), explains why the TFT trained from scratch improves from 0.891 pp with the earlier inputs to 0.401 pp with the shared inputs: dropping the 17 diagnostic features lowers the error by 0.17–0.35 pp and adding the calendar time between checks by 0.19–0.31 pp, both replicated at both learning rates ($p_{\mathrm{Holm}} \leq 0.002$). The learning rate has no significant effect in either seed set.

What the TFT uses. Its top-weighted input changes between seeds, inputs constant within S6 receive weight, and its attention over the four previous origins is nearly uniform (0.24–0.26 per lag): the weights are not a reliable measure of importance here, and the TFT makes little use of its history.

E. Learned residual on linear extrapolation

Table V and Fig. 5 give the results of the residual form (6). The criterion fixed in advance for a positive result, residual

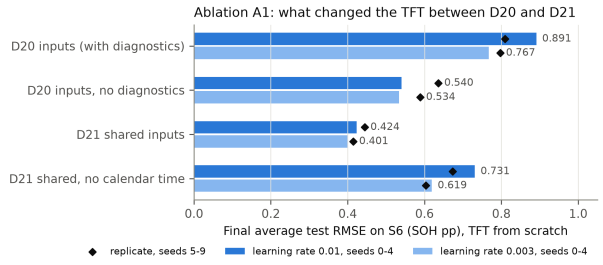


Fig. 4. Post hoc ablation of the TFT trained from scratch on S6: final average test RMSE for four input sets and two learning rates. Bars: mean of 50 runs on seeds 0–4; diamonds: replicate seeds 5–9.

TABLE V
LEARNED RESIDUAL ON S6 (STREAM FROM SCRATCH): FINAL AVERAGE TEST RMSE (PP), MEAN OVER 50 RUNS, MAIN / REPLICATE SEED SETS, AND FORGETTING (PP) FOR HORIZONS 1–6.

Method	E1: h=1–3	E2: h=1–6	Forgetting, E2
TFT + EWC, residual	**0.339 / 0.314**	**0.561 / 0.534**	**0.093 / 0.086**
TFT fine-tuning, residual	0.347 / 0.337	0.673 / 0.612	0.282 / 0.220
TFT + EWC, direct	0.504 / 0.487	0.802 / 0.835	0.135 / 0.166
MLP + EWC, residual	0.374 / 0.366	0.659 / 0.694	–
Linear extrapolation	0.343	0.485	–
Gradient boosting	0.451	0.898	–

TFT + EWC beating linear extrapolation, was *not* met: the two are practically equivalent for horizons 1–3 (differences of -0.004 and -0.029 pp), and for horizons 1–6 extrapolation is better on average ($+0.076$ and $+0.049$ pp, not replicated). Per horizon, extrapolation is better at $h = 1$ and 2 (replicated), and the gap closes without reversing by $h = 5$–6. Learning the residual is, however, what makes the TFT competitive: it improves TFT + EWC over the direct form by 0.17 pp (E1) and 0.24–0.30 pp (E2), and puts it ahead of gradient boosting, all replicated; it is the most accurate learned model on the S6 stream. With horizons 1–6, where the residual TFT forgets more, EWC lowers its forgetting from 0.28/0.22 to 0.09 pp and its backward transfer from 0.18/0.14 to 0.03 pp, replicated; its effect on the final error is not. For horizons 1–3, EWC and fine-tuning are practically equivalent.

Ageing-cycle inputs. Adding the 21 Spark-extracted inputs (Section III-B) to the residual TFT + EWC, as specified before the run, makes it worse: 0.480/0.501 against 0.339/0.314 pp, and worse than linear extrapolation, both replicated; gradient boosting also worsens (0.495 against 0.451 pp).

V. DISCUSSION

Why the proposed model does not win. TFT + EWC never beat linear extrapolation. Fade over one to three checks is smooth, so extrapolating the recent fade rate leaves little structure for a sequence model to add; the groups are few and small (21–27 and 135–225 training samples per task in S1 and S6); and the TFT barely uses its history window. EWC behaves as expected where there is something to protect: forgetting is negligible across the near-identical S1 cells and substantial

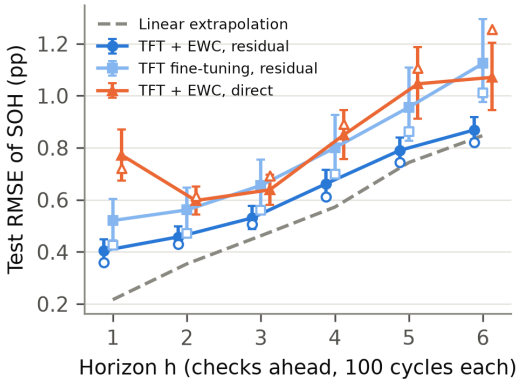


Fig. 5. E2: test RMSE by horizon after the last stream step on S6. Filled markers: mean and 95% bootstrap CI over 50 runs (seeds 0–4); open markers: replicate seeds 5–9. Dashed: linear extrapolation.

across the S6 usage conditions. Even there its benefit to the direct TFT did not survive a re-run, and after pretraining validation selected almost no EWC. Its clearest benefit came once the TFT learned only the correction to extrapolation: over six horizons, EWC cut that model's forgetting by about two thirds, replicated.

What matters. Building on the best simple forecaster mattered most: the learned residual made TFT + EWC the most accurate learned model on the S6 stream and level with extrapolation for horizons 1–3. The ablation also points to the inputs rather than the architecture: the diagnostic features hurt the TFT, and the calendar time between checks helped, consistent with fade per cycle rising with that time at 50 °C in all 20 S6 cells. Usage known in advance is a strong predictor, and forecasts are conditional on it. Dense measurements between checks did not help: with few cells per condition, 21 extra inputs added more variance than information. Pretraining on 369 first-life cells of other chemistries did not transfer to the LFP cells of S6, although retention of the pool was not the problem.

Reproducibility. Several significance results changed in a re-run on different hardware (Section IV-C); small datasets, many low-variance runs and Holm-corrected tests make such fragility likely. We recommend recording the full software environment, tuning with enough runs to separate neighbouring grid values, and requiring that claimed differences replicate on a second seed set and exceed a practically meaningful threshold.

Reuse and positioning. The reuse-class mapping is informative only when the forecast horizon crosses class boundaries: S1 stays in C0 and the S6 stream's test periods lie in C3, while on unseen S6 conditions the classes agree for 95–97% of samples with the learned models. The studies of Section II estimate the current SOH or SOC on other cells and with other SOH definitions, so their errors are not directly comparable with ours.

Limitations. The task was changed to forecasting after the

first dataset was inspected; the second-life datasets are small; S6 is aged at 50 °C and its "second life" continues the same protocol; SOH is relative to each cell's first valid measurement; forecasts are conditional on planned usage; and the ablation and the residual experiment were decided after the main results (the latter was specified and committed before it was run).

VI. Conclusion

We built a leakage-controlled benchmark for forecasting the SOH of second-life lithium-ion cells from five public datasets and evaluated a Temporal Fusion Transformer with Elastic Weight Consolidation as a continual learner. Learned forecasters beat persistence clearly, and forgetting appears whenever usage conditions differ. Learning the correction to linear extrapolation made TFT + EWC the most accurate learned model on the S6 stream, 0.17–0.30 pp better than its direct form and level with extrapolation for horizons 1–3, and with this form EWC reduced forgetting over longer horizons; all of these replicated on a second seed set. No model, however, beat extrapolation of the recent fade rate, first-life pretraining on other chemistries did not transfer, and inputs mattered more than architecture. Future work should test continual learning on larger and more varied second-life populations, use the dense per-cycle data between checks, and quantify forecast uncertainty near reuse-class boundaries.

References

[1] E. Braco, I. San Martín, P. Sanchis, A. Ursúa, and D.-I. Stroe, "State of health estimation of second-life lithium-ion batteries under real profile operation," *Applied Energy*, vol. 326, p. 119992, 2022.

[2] ——, "Health indicator selection for state of health estimation of second-life lithium-ion batteries under extended ageing," *Journal of Energy Storage*, vol. 55, p. 105366, 2022.

[3] M. Faraji-Niri, M. Rashid, J. Sansom, M. Sheikh, D. Widanage, and J. Marco, "Accelerated state of health estimation of second life lithium-ion batteries via electrochemical impedance spectroscopy tests and machine learning techniques," *Journal of Energy Storage*, vol. 58, p. 106295, 2023.

[4] A. Takahashi, A. Allam, and S. Onori, "Evaluating the feasibility of batteries for second-life applications using machine learning," *iScience*, vol. 26, p. 106547, 2023.

[5] X. Cui, M. A. Khan, G. Pozzato, S. Singh, R. Sharma, and S. Onori, "Taking second-life batteries from exhausted to empowered using experiments, data analysis, and health estimation," *Cell Reports Physical Science*, vol. 5, p. 101941, 2024.

[6] V. S, H. S. Che, J. Selvaraj, K. S. Tey, J. W. Lee, H. Shareef, and R. Errouissi, "State of health (SoH) estimation methods for second life lithium-ion battery—review and challenges," *Applied Energy*, vol. 369, p. 123542, 2024.

[7] M. Lee, S. Kim, S. Kim, and J.-I. Choi, "Bilevel-optimized continual learning for predicting capacity degradation of lithium-ion batteries," *Journal of Energy Storage*, vol. 86, p. 111187, 2024.

[8] J. Yao, B. Zheng, and J. Kowal, "Continual learning for online state of charge estimation across diverse lithium-ion batteries," *Journal of Energy Storage*, vol. 117, p. 116086, 2025.

[9] X. Li, Y. Lin, S. Yang, and W. Zhang, "Intelligent domain-generalized second-life EV battery state-of-health estimation," *Journal of Energy Storage*, vol. 140, p. 118989, 2025.

[10] S. Meskine, S. Al-Majeed, and H. El Asri, "Reliability assessment of second-life EV batteries using probabilistic deep learning models for state-of-health prediction," *World Electric Vehicle Journal*, vol. 17, p. 441, 2026.

[11] K. Moy, M. A. Khan, S. Fasolato, G. Pozzato, A. Allam, and S. Onori, "Second-life lithium-ion battery aging dataset based on grid storage cycling," *Data in Brief*, vol. 57, p. 111046, Dec. 2024.

[12] ——, "Second-life lithium-ion battery aging dataset based on grid storage cycling," Open Science Framework, dataset, file version 2 (27 Feb. 2025), 2024, licensed under CC BY 4.0. Accessed 29 Sep. 2026. [Online]. Available: https://doi.org/10.17605/OSF.IO/8JNR5

[13] W. Wheeler, Y. Bultel, P. Venet, and A. Sari, "An ageing study of twenty 18650 lithium-ion graphite/LFP cells in first and second life use," *Scientific Data*, vol. 12, p. 392, 2025.

[14] W. Wheeler, P. Venet, Y. Bultel, and A. Sari, "Aging study on twenty A123 18650 graphite/LFP 1.1 Ah cells," Recherche Data Gouv, dataset, version 2.0 (11 Mar. 2025), 2025, licensed under CC BY 4.0. Accessed 29 Sep. 2026. [Online]. Available: https://doi.org/10.57745/OLBXKT

[15] M. Luh and T. Blank, "Comprehensive battery aging dataset: capacity and impedance fade measurements of a lithium-ion NMC/C-SiO cell," *Scientific Data*, vol. 11, p. 1004, 2024.

[16] ——, "Comprehensive battery aging dataset: capacity and impedance fade measurements of a lithium-ion NMC/C-SiO cell [dataset – version 2: result data]," Karlsruhe Institute of Technology, RADAR4KIT, 2024, cC BY 4.0.

[17] F. Wang, Z. Zhai, Z. Zhao, Y. Di, and X. Chen, "Physics-informed neural network for lithium-ion battery degradation stable modeling and prognosis," *Nature Communications*, vol. 15, p. 4332, 2024.

[18] F. Wang, "Project – physics-informed neural network for lithium-ion battery degradation stable modeling and prognosis," Zenodo, version V1, 2024, cC BY 4.0.

[19] J. Zhu, Y. Wang, Y. Huang, R. Bhushan Gopaluni, Y. Cao, M. Heere, M. J. Mühlbauer, L. Mereacre, H. Dai, X. Liu, A. Senyshyn, X. Wei, M. Knapp, and H. Ehrenberg, "Data-driven capacity estimation of commercial lithium-ion batteries from voltage relaxation," *Nature Communications*, vol. 13, p. 2261, 2022.

[20] ——, "Data-driven capacity estimation of commercial lithium-ion batteries from voltage relaxation [dataset]," Zenodo, 2022, cC BY 4.0.

[21] G. Pozzato, A. Allam, and S. Onori, "Lithium-ion battery aging dataset based on electric vehicle real-driving profiles," *Data in Brief*, vol. 41, p. 107995, Apr. 2022.

[22] K. Moy and S. Onori, "Synthetic grid storage duty cycles for second-life lithium-ion battery experiments," *SAE International Journal of Advances and Current Practices in Mobility*, vol. 6, no. 1, pp. 261–269, 2023.

[23] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.

[24] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[25] D. Rolnick, A. Ahuja, J. Schwarz, T. P. Lillicrap, and G. Wayne, "Experience replay for continual learning," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[26] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.

[27] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, 2015.

[28] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[29] A. Chaudhry, P. K. Dokania, T. Ajanthan, and P. H. S. Torr, "Riemannian walk for incremental learning: Understanding forgetting and intransigence," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018.

[30] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman & Hall, 1993.

[31] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[32] D. S. Kerby, "The simple difference formula: An approach to teaching nonparametric correlation," *Comprehensive Psychology*, vol. 3, p. 11.IT.3.1, 2014.

[33] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.